

SQL INSIGHTS

Game Analytics: Unlocking Tennis Data

Business questions and insights from the SportRadar API —
competitions, venues and doubles rankings



Python • SQL • Streamlit

Three datasets, twenty business questions

Q1–Q7

Competitions

Categories, types, genders and
parent–child hierarchies

Q8–Q14

Venues & Complexes

Courts, cities, countries and
timezones

Q15–Q20

Doubles Rankings

Competitor ranks, points,
movement and countries

What does our competition catalogue look like across categories?



83.8%

ITF Men + ITF Women share

KEY INSIGHTS

- 2,000 competitions mapped across 10 category names, dominated by lower-tier developmental tours.
- ITF Men (59.6%) and ITF Women (24.2%) make up the vast majority; ATP and WTA together are under 4%.
- Singles (49.9%) and doubles (50.1%) are evenly split, while men's events lead overall (72.2%).

How many competitions does each category hold?



63.2%

ITF Men + Women of 6,689

KEY INSIGHTS

- ITF Men (2,198) and ITF Women (2,032) dominate tournament volume, followed by Challenger at 15.9% (1,065).
- Elite main-tour categories (ATP and WTA) total just 7.2% (484 competitions).
- Team and exhibition events such as Davis Cup, Billie Jean King Cup and Hopman Cup are single standalone fixtures.

Which competitions are doubles events?



2,000

doubles competitions

KEY INSIGHTS

- Every record returned strictly belongs to the doubles event type.
- Men's events make up 55.95% (1,119) of doubles competitions.
- Women's events account for the remaining 44.05% (881).

What does a single category look like? (ITF Men)



2,000

ITF Men competitions

KEY INSIGHTS

- All 2,000 competitions filtered under the ITF Men category were retrieved.
- Event formats are almost perfectly balanced: 1,000 doubles and 999 singles.
- Events are spread across international circuit locations.

Which competitions have sub-events (parent–child hierarchy)?



1

parent–child link found

KEY INSIGHTS

- A single hierarchy exists: ITF Romania F9, Men Singles is the parent of ITF Romania F9, Men Doubles.
- The doubles event is configured as a secondary tournament under the primary singles competition.

How are singles and doubles distributed within each category?



18

mixed events (<0.3%)

KEY INSIGHTS

- ITF Men/Women, Challenger, WTA and ATP keep a near 1:1 balance between singles and doubles.
- UTR competitions run exclusively as singles formats.
- Mixed doubles are exceptionally rare and appear almost only in exhibition, ATP and team events (Hopman Cup, United Cup).

Which competitions stand alone as top-level events?



603

top-level (ROOT) events

KEY INSIGHTS

- These events operate independently, without a parent tournament.
- Universal Tennis Rating (UTR) events dominate at 92.5% (558 competitions).
- The other 7.5% are international team cups (Davis Cup, United Cup, Hopman Cup) and select exhibition or tour fixtures.

How are venues mapped to their tennis complexes?



342

unique complexes

KEY INSIGHTS

- 2,000 venue records are mapped across 342 unique complexes; generic labels like “Court 1” and “Court 2” are most frequent.
- National Tennis Center (48 venues) and Buenos Aires Lawn Tennis Club (28) hold the most court mappings.

Which complexes have the most courts?



4,124

venues across 778 complexes

KEY INSIGHTS

- Leaders: Buenos Aires Lawn Tennis Club (29), Melbourne Park (28) and Queensland Tennis Centre (25).
- Most active complexes have 3–6 courts (39.3% of complexes).
- 161 complexes (20.7%) have no mapped courts recorded — a data gap worth closing.

What venue infrastructure exists in a specific country? (Chile)



65

venues in Chile

KEY INSIGHTS

- Santiago hosts 30 of Chile's 65 venues (46.2%) across 7 cities; one complex holds nearly 30% of mapped courts.
- Names blend Spanish and English ("Cancha Central" vs "Centre Court"); all venues sit in the America/Santiago timezone.

Which timezones do our venues operate in?



77

unique timezones

KEY INSIGHTS

- 2,000 venues span 77 global timezones across major tennis regions.
- America/New_York (176) and Asia/Shanghai (167) lead, followed by Europe/Paris (158) and Europe/Rome (112).

Which complexes are true multi-court facilities?



589

multi-venue complexes

KEY INSIGHTS

- Together they hold 4,096 courts, averaging about 7 venues per complex (range 2–29).
- Buenos Aires Lawn Tennis Club (29), Melbourne Park (28) and Queensland Tennis Centre (25) lead on capacity.

Which countries host the most venues?



20.0%

of venues in top 3 nations

KEY INSIGHTS

- 2,000 venues across 75 countries, led by the USA (187, 9.4%), France (109) and Italy (104).
- Saudi Arabia (1), Israel (2) and the Philippines (3) sit at the low end — potential growth markets.

What venues belong to one complex? (Nacional)



1

court recorded

KEY INSIGHTS

- Only one venue is mapped to the Nacional complex: Cancha Central (sr:venue:70045).
- This shows the complex has a single recorded court facility in the dataset.

Who are the competitors, and what are their ranks and points?



11,350

top points (Katerina Siniakova)

KEY INSIGHTS

- 1,000 competitors show a clear inverse link between rank and points ($r = -0.63$); top-ranked players hold over 10,000 points.
- Katerina Siniakova leads with 11,350 points at Rank 1; the average across the pool is 725.6 points (ranks 1–500).

Who sits at the very top of the standings?



Top 5

led by three Rank 1 athletes

KEY INSIGHTS

- Harri Heliovaara, Katerina Siniakova and Henry Patten are all listed at world Rank 1 (separate doubles rankings).
- Taylor Townsend is Rank 2 and Neal Skupski Rank 3; ranks 4 and 5 have no entries in this subset.

Which competitors held a stable rank this week?



117

competitors with zero movement

KEY INSIGHTS

- These players maintained their rank with no net movement.
- The group includes top doubles specialists Harri Heliovaara, Henry Patten and Neal Skupski — high stability at the peak.

How strong is a specific country's ranking presence? (Croatia)



11,917

total Croatian ranking points

KEY INSIGHTS

- Croatian competitors have accumulated 11,917 combined ranking points.
- This reflects a strong collective performance within international standings.

Which countries have the most competitors?



20.6%

of players from top 3 countries

KEY INSIGHTS

- USA leads with 106 competitors (10.6%), followed by Great Britain (52) and France (48) across 73 nations.
- The top three account for 206 of the 1,000 players — notable geographic concentration.

Who has the highest points in the current week?



11,350

points — Rank 1

KEY INSIGHTS

- Katerina Siniakova of Czechia holds the highest total, in women's doubles.
- She is firmly at Rank 1, with a substantial lead at the top of the global standings.

Key takeaways

Developmental tours dominate

ITF Men and Women are the bulk of competition volume; ATP/WTa are a small elite slice.

Infrastructure is concentrated

The top 3 countries host 20% of venues, and 161 complexes have no mapped courts.

Doubles ranks are tightly held

Rank 1 is shared and 117 players held steady, led by Siniakova at 11,350 points.

USA leads representation

106 of 1,000 competitors come from the USA, ahead of Great Britain and France.